International Radiation Detectors, Inc.

Photodiode Selection

Photodiode users are requested to select the optimum devices for their applications as follows:

1) AXUV (Absolute XUV) series devices can be used to cover the complete photon spectral rage (0.04 nm to 1100 nm) and should be used for detection of low energy electrons and ions because of their 6 nm oxide window and 100% internalquantum efficiency.

2) Because the oxide window will create surface recombination (resulting in loss of 100% internal quantum efficiency) after receiving several G-rad (SiO2) dose, SXUV series diodes with metal silicide window have been developed. These diodes will have almost infinite radiation hardness to photons. Owing to this feature, SXUV series diodes are recommended when the high photon flux is involved and for detection of UV/EUV pulse radiation and when the pulse energy density is higher then 0.1 µJ/cm2.

3) Like the AXUV series diodes, UVG series diodes have 100% internal quantum efficiency. The oxide window on these devices is 40 to 150 nm thick which is optimized for 130 nm to 1100 nm photon detection.